Software Requirements Specification (SRS)

Bustard Beacon- Conservation System for Great Indian Bustard

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1. Introduction

1.1 Project Introduction

Bustard Beacon is a web-based system developed for monitoring and protecting the Great Indian Bustard. The system helps forest staff to report bird sightings, health problems, threats and other suspicious activities.

The system connects rangers, forest officers, medical teams, researchers and the administrator on one platform. A ranger can upload an image, description and location of an incident. The concerned forest officer receives the request, verifies it and takes the required action. Medical teams can also be contacted for health-related cases.

The system also stores bird movement, health and incident records. This data can later be used by researchers to study migration patterns and identify areas that need more attention.

1.2 Purpose of System

The main purpose of the system is to make bird monitoring and emergency reporting easier and faster.

The system will:

- Monitor bird sightings and movement.
- Report suspicious activities and threats.
- Send alerts to the concerned forest officer.
- Give priority to serious cases.
- Reduce manual record keeping.
- Help officers assign nearby rangers.
- Manage medical cases.
- Maintain bird and incident records.
- Help analyse migration and conservation data.

1.3 Scope of Project

The system will provide:

- Login for different users.
- Incident reporting with image and location.
- Real-time alerts.
- Incident priority management.
- Ranger and patrol team management.
- Migration analysis.

2. Existing System

At present, wildlife monitoring can involve manual records, phone calls and separate communication between field staff and officers. Because of this, important information may be delayed or difficult to track.

Some major problems are:

- Delay in reporting emergencies.
- Difficulty in contacting the correct officer.
- Manual maintenance of records.
- No single place for bird information.
- Difficulty in tracking bird movement.
- Manual preparation of reports.
- Difficulty in identifying high-risk areas.

3. Objectives

The objectives of the system are:

1. To monitor Great Indian Bustard sightings and movement.
2. To report threats and suspicious activities.
3. To send real-time alerts to forest officers.
4. To prioritise incidents according to their seriousness.
5. To reduce manual work in reporting and record keeping.
6. To help officers assign nearby rangers or patrol teams.
7. To maintain bird, health and incident history.
8. To manage restricted and sensitive tourism areas.
9. To analyse migration and other conservation data.

4. Overall Description

Bustard Beacon will have the following main modules:

Module	**Purpose**
Login	User authentication
Incident Management	Report and track incidents
Alert Management	Send notifications
Patrol Management	Manage patrol teams
Medical Management	Handle health cases
Tourism Management	Manage restricted areas
Research & Analysis	Analyse collected data
Reports	Generate required reports

5. User Roles

5.1 Admin

Admin manages the overall system.

Main functions:

- Add, update and delete users.
- Give or remove permissions.
- Manage bird information.
- View all requests.
- Manage system data.
- Generate overall reports.

5.2 Ranger

Ranger works in the field and reports bird-related activities.

Main functions:

- Login to the system.
- Report bird sightings.
- Upload image and description.
- Select issue category.
- Share current location.
- Report health or human-related threats.
- Check request status.

The system sends the request to the forest officer responsible for that zone.

5.3 Forest Officer

Forest officer verifies requests and manages the response.

Main functions:

- Receive and verify requests.
- Check incident details.
- Set priority.
- Assign nearby rangers or patrol teams.
- Manage patrol teams.
- Contact medical team when required.
- Manage restricted tourism areas.
- Prepare final reports.

5.4 Medical Team

Medical team handles health-related cases.

Main functions:

- Receive medical requests.
- Check bird condition.
- Record treatment and medical details.
- Maintain medical history.
- Record required equipment.
- Prepare medical reports.

For serious cases, the ranger can directly send the request to the medical team.

5.5 Researcher

Researcher uses the stored data for analysis.

Main functions:

- Analyse bird sightings.
- Analyse monthly and yearly incidents.
- Study migration patterns.
- Identify high-risk areas.
- Compare historical data.
- Suggest areas for improvement.

6. Functional Requirements

The Bustard Beacon system shall provide the following functional requirements to improve monitoring, conservation, and threat management of the Great Indian Bustard (GIB)**.**

FR-01 User Authentication

The system shall provide secure login access for different user roles, namely Administrator, Ranger, Forest Officer, Medical Team, and Researcher. Each user shall access only the features and information relevant to their responsibilities, ensuring secure handling of wildlife conservation data.

FR-02 User and Permission Management

The Administrator shall be able to create, update, deactivate, and delete user accounts. The Administrator shall also assign role-based permissions so that Rangers can report incidents, Forest Officers can validate and manage cases, Medical Teams can handle bird treatment, and Researchers can access analytical data.

FR-03 Great Indian Bustard Information Management

The system shall maintain a centralized database containing detailed information about Great Indian Bustards, including bird identification, sightings, health status, breeding records, movement history, habitat location, and previous incidents. This eliminates manual record keeping and provides a single source of information for conservation authorities.

FR-04 Bird Sighting and Threat Reporting

The Ranger shall be able to report new Great Indian Bustard sightings or threats directly from the field by entering the incident category, description, date, time, GPS location, and photographs. The system shall support reporting of threats such as illegal hunting, injured birds, habitat destruction, collisions with power lines, and suspicious human activities.

FR-05 Automatic Zone Identification

After an incident is reported, the system shall automatically identify the forest division or conservation zone based on the GPS location and forward the report to the responsible Forest Officer. This reduces delays caused by manual communication and ensures that reports reach the correct authority quickly.

FR-06 Real-Time Alert Management

The system shall immediately notify the concerned Forest Officer whenever a new incident or bird sighting is reported. If the reported incident involves an injured or sick Great Indian Bustard, the system shall simultaneously notify the Medical Team to enable faster rescue and treatment.

FR-07 Incident Verification and Validation

The Forest Officer shall review all submitted reports by examining the provided images, location, and description. The officer shall approve the report, reject invalid submissions, or request additional information from the reporting Ranger before initiating further action.

FR-08 Incident Priority Classification

The Forest Officer shall classify every validated incident according to its severity (Critical, High, Medium, or Low). Incidents involving injured birds, poaching, or immediate threats to the species shall receive higher priority so that rescue and protection activities can begin without delay.

FR-09 Ranger and Patrol Assignment

The Forest Officer shall assign the nearest available Ranger or Patrol Team to respond to validated incidents. The system shall track task assignments, current status, completion details, and response history, improving coordination among field personnel.

FR-10 Medical Case Management

The Medical Team shall record the health condition of injured Great Indian Bustards, prescribed treatment, medicines, equipment used, rehabilitation progress, and final medical

reports. The system shall maintain complete medical histories for future reference and conservation studies.

FR-11 Habitat and Tourism Zone Management
The Forest Officer shall maintain and update information about normal, restricted, breeding, and critical conservation zones. The system shall help prevent unnecessary tourist or human activities in sensitive habitats where Great Indian Bustards are vulnerable.

FR-12 Research and Conservation Analysis
Researchers shall analyse historical and real-time data related to bird sightings, migration routes, nesting locations, incidents, seasonal movement, mortality, and habitat conditions. The system shall support conservation planning by identifying high-risk locations and migration trends.

FR-13 Report Generation
The system shall automatically generate reports such as incident reports, rescue reports, patrol reports, medical reports, migration statistics, and conservation summaries. These reports shall assist Forest Departments in decision-making and policy planning.

FR-14 Activity Logging and Audit Trail
The system shall record important activities performed by all users, including login history, incident reporting, validation, patrol assignments, medical updates, and report generation. This ensures transparency, accountability, and future auditing of conservation activities.

7. Non-Functional Requirements

These requirements should also be connected with your conservation project instead of remaining generic.

Security
The system shall implement role-based access control so that only authorized users can access sensitive conservation data. Rangers shall only report incidents, Forest Officers shall validate and manage operations, Medical Teams shall access treatment records, Researchers shall access analytical data, and only the Administrator shall manage user accounts.

Performance
The system shall process incident reports and generate alerts within a few seconds to ensure rapid response to emergencies such as injured Great Indian Bustards, poaching attempts, or habitat threats.

Usability
The system shall provide a simple, mobile-friendly interface that enables Rangers to quickly submit reports with minimal typing while working in remote wildlife habitats, even under challenging field conditions.

Reliability
The system shall ensure that all bird records, incident reports, medical histories, patrol assignments, and research data are stored accurately without data loss, ensuring continuous availability for conservation activities.

Scalability
The system shall support future expansion to accommodate larger numbers of users, additional wildlife species, multiple wildlife sanctuaries, and increasing volumes of conservation data without affecting performance.

Maintainability
The system architecture shall be modular so that future enhancements such as AI-based bird identification, GIS mapping, GPS tracking, predictive migration analysis, and mobile applications can be integrated easily.

Backup and Recovery
The system shall perform regular automated database backups and provide recovery mechanisms to prevent loss of critical Great Indian Bustard monitoring and conservation data due to hardware failures or unexpected system errors.

8. System Workflow

- Ranger reports the incident with details, image and location.
- System identifies the zone and alerts the Forest Officer.
- Nearby Ranger/Patrol Team is assigned.
- Medical Team is involved if required.
- Action is taken and the case is closed.
- Data is stored for reports and research analysis.

9. Technology Stack

Technology	**Use**
HTML, CSS, Bootstrap	Frontend
Django / Python	Backend
MySQL	Database
JavaScript	Client-side functionality
Google Maps / OpenStreetMap	Location
Chart.js	Data analysis and charts
Git/GitHub	Version control

10. Future Scope

In future, the system can be extended with:

- Mobile application for rangers.
- GPS tracking for bird movement.
- AI-based bird/image identification.
- Offline reporting in areas with poor network.
- Machine learning for migration prediction.
- GIS-based map showing sightings and threat areas.

11. Conclusion

Bustard Beacon provides a common platform for monitoring and protecting the Great Indian Bustard. It connects rangers, forest officers, medical teams, researchers and administrators.

The system reduces manual work by providing digital reporting, alerts, incident tracking, patrol assignment and report generation. It also stores bird movement and health information, which can help in studying migration and improving conservation activities.

Software Requirements Specification
Bustard Beacon — v1.0

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1. Introduction
1.1 Purpose
The purpose of Bustard Beacon is to provide a centralized web-based monitoring and threat alert system for the conservation of the Great Indian Bustard (GIB). The system enables Rangers to report bird sightings, injuries, poaching incidents, habitat destruction, and other threats using GPS location and images. Forest Officers receive real-time alerts, verify incidents, assign patrol teams, and coordinate rescue operations. Medical Teams manage treatment records, while Researchers analyse migration patterns and conservation data. The system aims to reduce manual paperwork, improve emergency response, maintain centralized wildlife records, and support informed conservation decision-making.
1.2 Scope
Bustard Beacon is a web-based application developed for wildlife conservation authorities responsible for protecting the Great Indian Bustard. The system provides secure role-based login for Administrators, Rangers, Forest Officers, Medical Teams, and Researchers. It supports bird information management, incident reporting with GPS location and photographs, automatic forest zone identification, real-time alert notifications, incident verification, patrol assignment, medical case management, conservation zone management, migration analysis, report generation, and activity logging. The initial version is designed as a web application, while mobile applications, AI-based bird identification, GIS mapping, and offline reporting are considered future enhancements.
1.3 Intended Audience
Bustard Beacon is a centralized wildlife conservation platform that digitizes the monitoring and protection of the Great Indian Bustard. It replaces manual reporting methods with a secure online system that allows authorized personnel to report, verify, manage, and analyse bird-related incidents efficiently.
1.4 Assumptions & Dependencies
Assumptions:
• All users have valid system credentials.
• Rangers have GPS-enabled devices with internet connectivity while reporting incidents.
• Forest conservation zones are predefined in the system database.
• Users perform operations according to their assigned roles.
• Images uploaded are clear enough for verification.

Dependencies:
• Python 3.x
• Django Framework
• MySQL Database
• Google Maps API/OpenStreetMap
• Bootstrap
• JavaScript
• Chart.js

• Git/GitHub
• Internet connection

2. User Roles

Role	Description
Administrator	Manages users, permissions, system settings, bird records, and generates administrative reports.
Ranger	Reports Great Indian Bustard sightings, injured birds, poaching incidents, uploads images, shares GPS location, and tracks incident status.
Forest Officer	Verifies incidents, assigns priorities, allocates Rangers and Patrol Teams, manages conservation zones, and coordinates rescue operations.
Medical Team	Handles injured birds, records treatments, rehabilitation progress, equipment used, and maintains medical history.
Researcher	Analyses bird sightings, migration routes, incident trends, habitat conditions, and generates conservation insights.

3. Functional Requirements

#	Category	Requirement	Priority
FR-01	Authentication	Users shall securely log in according to their assigned roles (Administrator, Ranger, Forest Officer, Medical Team, and Researcher).	High
FR-02	Bird Management	The system shall maintain centralized records of Great Indian Bustard sightings, health status, movement history, and habitat information.	High
FR-03	Incident Reporting	Rangers shall report bird sightings, injuries, poaching, habitat destruction, and other threats using GPS location and photographs.	High
FR-04	Alerts	The system shall automatically identify the responsible forest zone and notify the concerned Forest Officer immediately after an incident is reported.	High
FR-05	Incident Management	Forest Officers shall validate incidents, assign priorities, and allocate nearby Rangers or Patrol Teams for appropriate action.	High
FR-06	Medical Management	Medical Teams shall record treatment details, rehabilitation progress, medicines, and medical reports for injured birds.	Medium
FR-07	Research	Researchers shall analyse migration routes, bird movement, incident history, and habitat conditions for conservation planning.	Medium
FR-08	Reporting	The system shall generate incident reports, medical reports, patrol reports, and conservation summaries.	Medium

4. Non-Functional Requirements

#	Category	Requirement	Priority
NFR-01	Security	Role-based access control shall restrict users to only the functions permitted by their assigned roles.	High
NFR-02	Performance	Incident reports and emergency alerts shall be processed within a few seconds.	High || NFR-03	Reliability	The system shall prevent data loss by maintaining reliable storage and regular database backups.	Medium

| NFR-04 | Scalability | The system shall support future expansion to additional wildlife species and conservation regions. | Low |

5. Use Cases

UC-1: Report a Bird Sighting or Threat
- **Actor:** Ranger
- **Flow:** 1. Ranger logs into the system.
2. Selects Report Incident.
3. Enters incident details.
4. Uploads bird image.
5. Shares GPS location.
6. Submits the report.
7. System stores the report and alerts the concerned Forest Officer.

UC-2: Validate and Assign Incident
- **Actor:** Forest Officer
- **Flow:** 1. Forest Officer receives notification.
2. Reviews incident details and images.
3. Verifies the report.
4. Assigns incident priority.
5. Allocates nearby Ranger or Patrol Team.
6. Updates incident status.

UC-3: Manage Bird Treatment
- **Actor:** Medical Team
- **Flow:** 1. Receives medical alert.
2. Examines injured bird.
3. Records treatment details.
4. Updates rehabilitation progress.
5. Generates medical report.

UC-4: Analyse Conservation Data
- **Actor:** Researcher
- **Flow:** 1. Logs into the system.
2. Selects analysis period.
3. Views migration and incident data.
4. Generates charts and reports.
5. Downloads analytical report.

UC-5: Manage Users and Permissions
- **Actor:** Administrator
- **Flow:** 1. Administrator logs in.
2. Opens User Management.
3. Creates or updates user accounts.
4. Assigns user roles and permissions.
5. Saves changes.